

TEACHER'S APPARATUS LIST - 2021 H2 Biology Prelim Paper 4
PREPARATION LIST (FOR QUESTION 1)

Q1	Apparatus / Reagents/ Chemicals	Quantity per candidate	Remarks
1	Plastic container labelled M1 , containing Milk solution M1 at room temperature.	At least 100 cm ³	
2	Plastic container labelled X , containing sample solution X at room temperature.	At least 20 cm ³	
3	Plastic container labelled C , containing 10% calcium chloride solution at room temperature. NOTE: C is an irritant.	At least 10 cm ³	
4	Plastic container labelled E , containing 0.25% Rennin enzyme solution at room temperature. NOTE: E is an irritant.	At least 10 cm ³	
5	Plastic container labelled DW , containing distilled water.	At least 100 cm ³	
6	500ml Beaker labelled Hot Water , for collecting hot water	1	
7	500ml Beaker labelled Tap Water , for collecting tap water	1	
8	500ml Beaker for setting up water bath	1	
9	Plastic vials (B197) or can be B43, capacity approx. 30 cm ³	5	
10	Thermometer	1	
11	Black card or paper, approximately 50mm x 50mm	1	
12	10.0 cm ³ syringe	1	
13	5.0 cm ³ syringe	2	Also used in Q2
14	1.0 cm ³ syringe	2	Also used in Q2
15	Test tube rack	1	Also used in Q2
16	Test tubes	7	Also used in Q2
17	Glass rod	1	Also used in Q2
18	Marker pen	1	Also used in Q2
19	Paper towel	4	Also used in Q2
20	Digital stopwatch	1	Also used in Q2
21	Beaker of distilled water, labelled for washing	1	Also used in Q2
22	Beaker, labelled waste	1	Also used in Q2
23	Suitable eye protection	1	
24	Clean dry cloth (e.g. for cleaning)	1	Also used in Q2
-	Source of Hot water at 55°C	-	At teacher's bench

At teacher's bench: Two hot water dispensers, each set to 55°C

Preparation of Reagents for Q1:

M1, X and C, can be prepared **the day before** the examination and put in covered containers. **M1** and **X** should be kept in a refrigerator.

The solutions must be at room temperature for the examination.

(i) M1, 30 cm³ of fresh full fat milk (= 100%)

Any source of fresh full fat milk may be used. UHT milk is also acceptable.

(ii) X, 10 cm³ of fresh full fat milk (but at 60%)

This is prepared by mixing 60 cm³ of **M1** with 40 cm³ of distilled water. Mix well to form 100 cm³.

- This is sufficient for 10 candidates.

(iii) C, 10 cm³ of 10% calcium chloride solution

This is prepared by putting 10 g of calcium chloride-6-water into 70 cm³ of distilled water and stirring continuously. Make up to 100 cm³ with distilled water.

- This is sufficient for 10 candidates.

E should be prepared **no more than 2 hours prior** to the examination.

(iv) E, 10 cm³ of **0.25%** Rennin solution

This is prepared by dissolving **0.25 g** of rennin powder in 100 cm³ of distilled water. Mix well.

- This is sufficient for 10 candidates.

PREPARATION LIST (FOR QUESTION 2)

Q2	Apparatus / Reagents/ Chemicals	Quantity per candidate	Remarks
1*	Light microscope, with 1 eyepiece graticule	1	Shared between 2 candidates. (Each has access for uninterrupted period of 1 hour 15min only)
2*	Stage micrometer (0.01mm), In a large plastic container, labelled M	1	
3	1 leaf specimen in a covered Petri dish labelled L (Neem Leaf – For stomata printing)	1	
4	Bottle of clear Nail Varnish/polish	1	
5	Microscope slide and coverslip	2	
6	White tile	1	
7	Mounting needle	1	
8	Forceps	1	
9	Scalpel	1	
10	Plastic container with 5% glucose solution, labelled Y Note: The identity of this item is described differently in the question paper as “ <i>Plastic container with 5% sucrose solution, labelled Y</i> ”. This is intentional. Candidates should not be informed.	At least 10 cm ³	
11	Plastic container with 0.5% glucose solution, labelled Z Note: The identity of this item is described differently in the question paper as “ <i>Plastic container with honey sample solution, labelled Z</i> ”. This is intentional. Candidates should not be informed.	At least 10 cm ³	
12	Plastic container with 0.5% glucose solution, labelled I Note: The identity of this item is described differently in the question paper as “ <i>Plastic container with 1% Invertase solution, labelled I</i> ”. This is intentional. Candidates should not be informed.	At least 10 cm ³	
13	Benedict’s solution (in bottle)	At least 25 cm ³	
14	Test tube holder	1	
15	Bunsen burner, wire gauze, tripod stand, lighter	1 of each	
16	Plastic dropper	2	
17	1 cm ³ syringe	1	Also used in Q1
18	5 cm ³ syringe	1	Also used in Q1
19	Test tubes	2	Also used in Q1
20	Glass rod	1	Also used in Q1
21	Marker Pen	1	Also used in Q1

22	Stopwatch	1	Also used in Q1
25	Paper towels	4	Also used in Q1

*Shared apparatus (Q2 no. **1 – 2**) will be passed over to 2nd candidate at 1h 15 min time point.

Preparation of Reagents for Q2:

Y, Z and I can be prepared **the day before** the examination and put in covered containers.
Y, Z and I should be kept in a refrigerator.

The solutions must be at room temperature for the examination.

(i) Y, 10 cm³ of **5.0%** glucose solution

This is prepared by sprinkling **5.0 g** of glucose, a little at a time, onto the surface of 50 cm³ of distilled water stirring continuously as you sprinkle. Make up to 100 cm³ with distilled water.

➤ This is sufficient for 10 candidates.

(ii) Z and I, 10 cm³ of **0.5%** glucose solution

This is prepared by sprinkling **0.5 g** of glucose, a little at a time, onto the surface of 50 cm³ of distilled water stirring continuously as you sprinkle. Make up to 100 cm³ with distilled water.

➤ This is sufficient for 10 candidates.

~ END OF PREP LIST ~